IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

COMPUTATIONAL SENSING AND IMAGING

Monday, 19 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are ONE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : A. Bhandari
Second Marker(s) : Sen Wang

© Imperial College London

Imperial College London

Department of Electronic and Electrical Engineering

Examination 2025

Computational Sensing and Imaging

MSc and EEE/EIE PART IV

May, 2025 • Exam Duration: Three Hours

All Questions are Compulsory

Where unavoidable, make reasonable assumptions for your solutions.
Explain your assumptions in the answer script.

Markers: Ayush Bhandari and Sen Wang.

Questions [100 Marks]

Q1 Plenoptic Functions in Imaging.

(a) State the mathematical definition of the Plenoptic Function. **[1 Mark]**

(b) What is the dimensionality of the Plenoptic Function. **[1 Mark]**

(c) Identify the different variables of the function. **[2 Marks]**

(d) Map each parameter in the plenoptic function (a) to an imaging modality. **[3 Marks]**

(e) Explain (mathematically) the creation of a 2D photograph from a plenoptic function. **[2 Marks]**

Q2 Linear System Theory.

(a) Let $\mathcal{L}$ be a linear, time-invariant system.
Show that the response of this system to input signal $f(t)$ is given by,

$$f(t) \xrightarrow{\text{Input}} \boxed{\mathcal{L}} \xrightarrow{\text{Output}} \mathcal{L}[f](t) = (f * g)(t)$$

where $g(t) = \mathcal{L}[\delta](t)$ denotes the impulse response. **[4 Marks]**

(b) Show that $f_\omega(t) = e^{\jmath\omega t}$ is the eigenfunction for the system $\mathcal{L}$, satisfying that

$$f_\omega(t) \xrightarrow{\text{Input}} \boxed{\mathcal{L}} \xrightarrow{\text{Output}} \mathcal{L}[f_\omega](t) = \beta_\omega f_\omega(t).$$

Compute the corresponding eigenvalue β_ω. **[4 Marks]**

Q3 Prony's Method for Frequency Estimation.

Consider sum-of-sinusoids (SoS) samples defined as

$$x[n] = \sum_{k=0}^{K-1} c_k e^{\jmath(\omega_k n + \varphi_k)}, \quad n = 0, \cdots, N-1. \tag{1}$$

Our goal is to estimate the sinusoidal parameters $\{c_k, \omega_k, \varphi_k\},\ k = 0, \cdots, K-1$.

(a) Let $h[n], n = 0, \cdots, K$ denote the filter with z-transform,
$H(z) = \prod_{k=0}^{K-1}(1 - e^{\jmath\omega_k} z^{-1})$. Show that,

$$(h * x)[n] = 0.$$

[4 Marks]

(b) Re-organize the annihilation equation $h[n], n = 0, \cdots, K$ in vector-matrix form as $\mathbf{Ah} = \mathbf{0}$ and specify the dimension of $\mathbf{A}$. **[4 Marks]**

(c) To estimate $\mathbf{h}$, what is the condition on the number of samples N? **[1 Mark]**

(d) Once $\mathbf{h}$ is known, how to determine $\{c_k, \omega_k, \varphi_k\},\ k = 0, \cdots, K-1$? **[3 Marks]**

Q4 Lens Equation.

Let $\{r_1, r_2\}$ with $|r_1| < |r_2|$ denote the roots of the quadratic equation below

$$R^2 - \beta R + \beta f(1-\rho) = 0,\ \beta \neq 0.$$

Show that, $\{-r_1, r_2\}$ satisfy the Lens Maker's equation,

$$(\rho - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \frac{1}{f}.$$

[4 Marks]

Q5 Constrained Least Squares (CLS).

Consider the following CLS problem:

$$\min_{\mathbf{x}} \|\mathbf{WHx} - \mathbf{y}\|_2^2, \quad \text{such that} \quad \begin{cases} \mathbf{C_1 x} = \mathbf{b}_1 \\ \mathbf{C_2 x} = \mathbf{b_2} \end{cases}. \tag{2}$$

(a) Write down the Lagrangian function. **[2 Marks]**

(b) Compute the optimality conditions.
Hint: You can use: $\partial_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{b}\|_2^2 = 2\mathbf{A}^T(\mathbf{Ax} - \mathbf{b})$. **[2 Marks]**

(c) Show that the Lagrangian multiplier satisfies that

$$\boldsymbol{\lambda} = 2\left(\mathbf{C}\left(\mathbf{A}^\top\mathbf{A}\right)^{-1}\mathbf{C}^\top\right)^{-1}\left(\mathbf{C}\left(\mathbf{A}^\top\mathbf{A}\right)^{-1}\mathbf{A}^\top\mathbf{y} - \mathbf{b}\right)$$

where $\mathbf{A} = \mathbf{WH}$, $\mathbf{C} = \begin{bmatrix}\mathbf{C}_1 \\ \mathbf{C}_2\end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix}\mathbf{b}_1 \\ \mathbf{b}_2\end{bmatrix}$. **[4 Marks]**

(d) Substituting the result of (c) in (b), show the optimal solution is:

$$\mathbf{x}^* = (\mathbf{A}^\top\mathbf{A})^{-1}\left(\mathbf{A}^\top\mathbf{y} - \mathbf{C}^\top(\mathbf{C}(\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{C}^\top)^{-1}(\mathbf{C}(\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\mathbf{y} - \mathbf{b})\right)$$

[2 Marks]

Q6 Fluorescence Lifetime Imaging with Depth Sensors.

Consider a fluorescent sample with a lifetime parameter λ_0. The measurements resulting from the interaction of a continuous wave with the sample can be modeled as,

$$m(t) = \delta(t) * \rho_0 e^{-\frac{t}{\lambda_0}} u(t),$$

where m represents the measured signal, δ denotes the Dirac distribution, and u is the Heaviside step function. In applications such as microscopy, the sample is typically placed at a distance d_0 from the imaging plane. Consequently, the scene response function (SRF) can be expressed as

$$h_{\mathsf{FLI}}(t) = \underbrace{\Gamma_0 \delta(t - t_0)}_{\text{Direct Reflection}} + \underbrace{\rho_0 e^{-\frac{t-t_0}{\lambda_0}} u(t - t_0)}_{\text{Fluorescent Sample}}, \quad t_0 = 2\frac{d_0}{c}.$$

(a) What is the Fourier Transform of the scene response function $h_{\mathsf{FLI}}(t)$? **[4 Marks]**

(b) Consider the case of continuous-wave time-of-flight imaging, where the probing signal is defined as $p(t) = 1 + p_0 \cos(\omega t)$ and a lock-in sensor is employed.

Show that the measurements are given by

$$m(t) = \left|\widehat{h}_{\mathsf{FLI}}(0)\right| + \left|\widehat{h}_{\mathsf{FLI}}(\omega)\right| \frac{p_0^2}{2} \cos\left(\omega t - \angle \widehat{h}_{\mathsf{FLI}}(\omega)\right),$$

where $\widehat{h}_{\mathsf{FLI}}(\omega)$ is the Fourier Transform of the SRF computed in the above question.

[5 Marks]

Q7 (a) Consider the weighted regularization problem below:

$$\min_{\mathbf{x}} \|\mathbf{W}\mathbf{x} - \mathbf{y}\|_2^2 + \lambda \|\mathbf{A}\mathbf{x}\|_2^2. \tag{3}$$

Compute the optimal solution.

Hint: You can use: $\partial_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 = 2\mathbf{A}^T(\mathbf{A}\mathbf{x} - \mathbf{b})$. **[5 Marks]**

(b) In the frequency estimation problem in Eq. (1) of Q3, the measurements are usually noisy. Hence, the estimation of $\mathbf{h}$ in Q3 (b) is formulated as

$$\min_{\mathbf{h}} \|\mathbf{A}\mathbf{h}\|_2^2, \quad \text{subject to} \quad \|\mathbf{h}\|_2 = 1. \tag{4}$$

Utilizing the result in (a), show that the optimal solution to the minimization above is the eigenvector of $\mathbf{A}^\top\mathbf{A}$ associated with the minimal eigenvalue. **[5 Marks]**

Q8 **Sparse Recovery and Soft-Thresholding.**

Consider the following recovery problem:

Let $\mathbf{x} \in \mathbb{R}^N$ be a sparse sequence with M nonzero elements. The observed measurements are characterized as $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{w}$ where $\mathbf{A}$ is certain orthogonal matrix and $\mathbf{w}$ is the vector of noise.

(a) Write down the least squares solution. **[1 Marks]**

(b) Outline the issues of the least square solution. **[2 Marks]**

(c) Write down the optimization formulation that leverages the sparsity of $\mathbf{x}$.

Hint: Use the ℓ_0-norm. **[2 Marks]**

(d) Describe the main steps of the Basis Pursuit algorithm that solves (c). **[5 Marks]**

(e) What are the potential issues of the optimization scheme in (c) and the approach in (d)? **[3 Marks]**

(f) Due to the limitations, the problem formulation in (c) is generally relaxed. Using the regularization strategy, write down the relaxed optimization problem.

Hint: Replace the ℓ_0-norm in (c) with ℓ_1-norm. **[2 Marks]**

(g) Let $\mathbf{A} = \mathbf{Id}$, write down the solution to the minimization problem in (f).

Hint: You can use the soft–thresholding operator. **[4 Marks]**

Q9 **One-Bit Time-Resolved Imaging.**

Given a bandlimited pulse $p(t)$ and its samples are $p[n] = p(nT)$ where T is the sampling period.

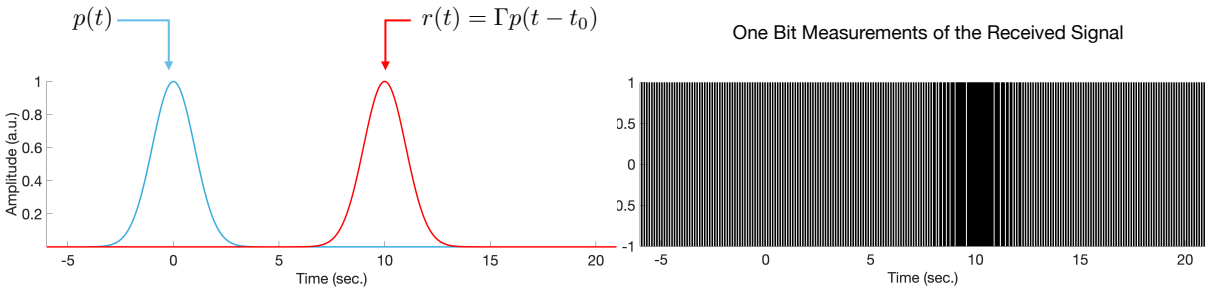


Fig. 1: Illustration of the one-bit time-resolved imaging.

(a) Outline the difficulty of achieving mega-pixel resolution for existing time-resolved imaging. **[2 Marks]**

(b) Due to this limitation, we consider combining the one-bit sampling with time-resolved imaging. In one-bit sampling, the samples $q[n] \in \{-1, 1\}$ are given by,

$$\begin{cases} q[n] &= \operatorname{sgn}(u[n-1] + f[n]) \\ u[n] &= u[n-1] + f[n] - q[n] \end{cases}.$$

Based on the illustration in Fig. 1, write down the one-bit samples of the reflected signal $r(t)$ as a single equation in terms of p and u. **[2 Marks]**

(c) From Q9(b) above, derive an equivalent expression in the Fourier domain. **[2 Marks]**

(d) To find the unknown time delay t_0, a common approach is to utilize *matched filtering*. In this way, one can estimate t_0 by solving the following maximization

$$t_0^\star = \arg\max_t |(r * \overline{p})(t)|, \quad \overline{p}(t) = p(-t). \tag{5}$$

Show that t_0 can be found by also utilizing match filtering for the one-bit samples q.

Hint: Utilize the fact that matched filtering with pulse p is equivalent to low-pass filtering.

[4 Marks]

Q10 Sparse-Deconvolution for Multi-Depth Imaging.

As illustrated in Fig. 2,

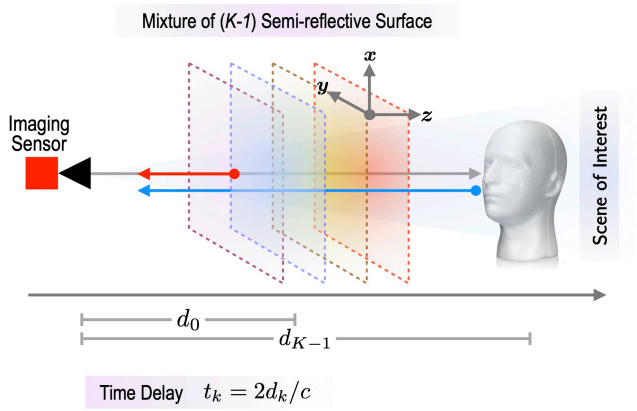


Fig. 2: Multi-Depth Imaging.

in multi-depth imaging, the measurements are given by

$$m(t) = \sum_{k=0}^{K-1} c_k \delta(t - t_k) * \varphi(t) = \sum_{k=0}^{K-1} c_k \varphi(t - t_k), \quad t = nT \tag{6}$$

where T is the sampling period and $\varphi(t)$ is the *known* Ω-bandlimited probing function.

Our goal is to find the scene parameters $\{c_k, t_k\}, k = 0, \cdots, K-1$.

(a) Match filtering (see Q9 (d)) could be utilized to find the scene parameters. Describe the potential difficulty in estimating the scene using match filtering. **[2 Marks]**

(b) Write the Fourier transform of (6). **[2 Marks]**

(c) Show that the scene parameters can be found using the Prony's method (see (1)). **[3 Marks]**

Q11 What is the difference between RGB image and hyperspectral image? **[2 Marks]**